A Distinguishing Memorization Using Hidden-Distribution Features

§5.2 shows differences in our extracted hidden-distribution features, namely the rank and probability across layers, between memorized idioms and non-memorized sequences. To verify that these features are distinguishing between memorized and non-memorized predictions, we build a classifier that receives them as input, as follows.

Experiment To answer the above, we represent every instance in IDIOMEM, for every LM we experiment with, as a sequence of probabilities and ranks assigned to the predicted token at each layer. This results in a feature vector for each instance in IDIOMEM for each of our LMs (GPT2_M, GPT2_L, BERT_B, and BERT_L). We then append a class label for each LM's instances corresponding to whether it memorized them. Then, for each LM's dataset, we perform 10-fold cross-validation with an 80%-20% train-test split to evaluate the accuracy of a logistic-regression classifier using the LogisticRegresion classifier of scikit-learn,[7] specifying L1 penalty and a bilinear solver as constructor parameters. After each split and before evaluating the classifier, we balance the test set by replacing the larger class with a random subsample the size of the smaller class. To isolate the distinguishing utility of ranks from that of probabilities, we repeat this process while only taking either of them as features at each time. We also repeat this process while separately considering just the last-layer probability as a single feature, the last hidden state vector, and finally, the ranks in layers of layers 1-12 (omitting ranks in layers 13-16 where ranks are usually 0) appended to all layer probabilities.

Results Results are given in Tab. 5. We observe that, across all models, most classifiers perform well over the 50%-accuracy baseline for their class-balanced test sets. The output probability alone is often highly distinguishing with around 78% accuracy, but the vector of 16 hidden-distribution probabilities seems to contain additional distinguishing information, as using it alongside ranks results in higher accuracy, typically around 84%. Using the ranks in addition to probabilities usually *decreases* accuracy, but omitting the ranks in layers 13-16 (which we know are mostly 0 as this is the confidence-boosting phase) attenuates this effect.

[7] https://scikit-learn.org/

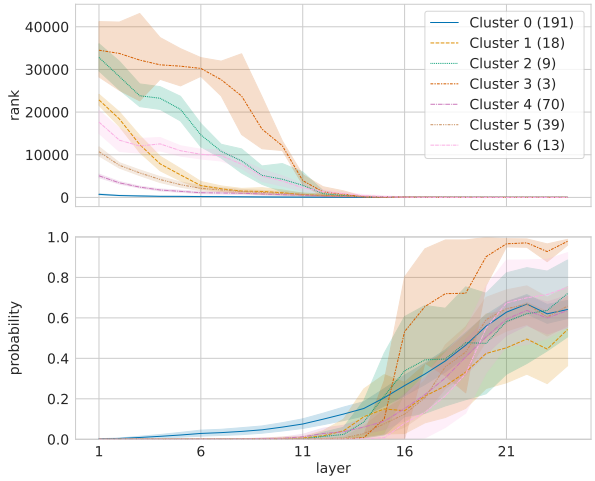

Figure 5: The predicted token's probability and rank across layers for each cluster, after clustering the memorized idioms in GPT2_M according to rank and probability across layers.

We conjecture that ranks have little meaningful information, especially in the confidence-boosting phase.

B Fine-grained Analysis of Memorized Predictions

§5.2 shows how recall of memorized predictions exhibits two characteristic phases (specifically, see Fig. 2). To shed light on this phenomenon, we conduct additional analysis.

B.1 Memorized Idioms In-depth Breakdown

In our analysis, we address all memorized predictions jointly. We now check whether these averaged results are consistent across subgroups of the memorized idioms. To this end, we cluster the memorized idioms by GPT2_M, using the same hidden features as in §A, i.e., each instance is represented by the predicted token probability and rank across layers. We then cluster the idioms into seven groups, using K-mean clustering, and visualize the prediction rank and probability across layers for each group. We set the number of clusters to $k = 7$ based on manual inspection, and as we observed no substantial differences in the resulting clusters for larger values of k.

Results are shown in Fig. 5. We find that all groups exhibit the confidence-boosting phase, as the prediction probability quickly increases starting from the intermediate layers. Notably, the lowest final probability observed is > 0.5, which is substantially higher than the average probability of

	$GPT2_M$	$GPT2_L$	$BERT_B$	$BERT_L$
probability	$\mathbf{84.6 \pm 2.8}$	$\mathbf{84.0 \pm 2.3}$	$\mathbf{84.2 \pm 1.8}$	$\mathbf{77.6 \pm 3.3}$
ranks	63.2 ± 1.6	59.5 ± 2.7	73.3 ± 3.3	64.5 ± 3.6
probability + ranks	83.6 ± 2.7	82.9 ± 2.9	82.7 ± 2.2	76.9 ± 2.9
ranks layer 1-12 + probability	$\mathbf{84.3 \pm 2.8}$	83.4 ± 2.7	82.7 ± 2.2	$\mathbf{77.2 \pm 4.2}$
probability last layer	79.5 ± 3.5	83.3 ± 2.3	77.7 ± 2.3	74.2 ± 4.8
final hidden state	72 ± 2.1	72.5 ± 3.7	72.7 ± 2.9	66.0 ± 3.0
token ids	58 ± 0.2	59.4 ± 0.2	58.8 ± 0.2	71.6 ± 0.2
random	49.5 ± 3.8	49.7 ± 4.7	50.4 ± 4.9	50.1 ± 3.5

Table 5: Cross-validation accuracy of a logistic-regression classifier trained to distinguish between memorized and non-memorized idioms.

~ 0.2 for non-memorized predictions (§5.2). However, considering the prediction rank for the different groups, we observe a relatively large variation. Specifically, we observe that 55% of the instances (cluster 0) have a low initial rank. This further supports the findings in §A.

B.2 Log-Scale Visualization

Fig. 6 shows a log-scaled view of the graphs from Fig. 2. We observe that, in terms of orders of magnitude (vs. absolute value), the differences in initial ranks between memorized and non-memorized predictions are more minor, especially in BERT models, whereas rank differences measured in upper layers are more stark.

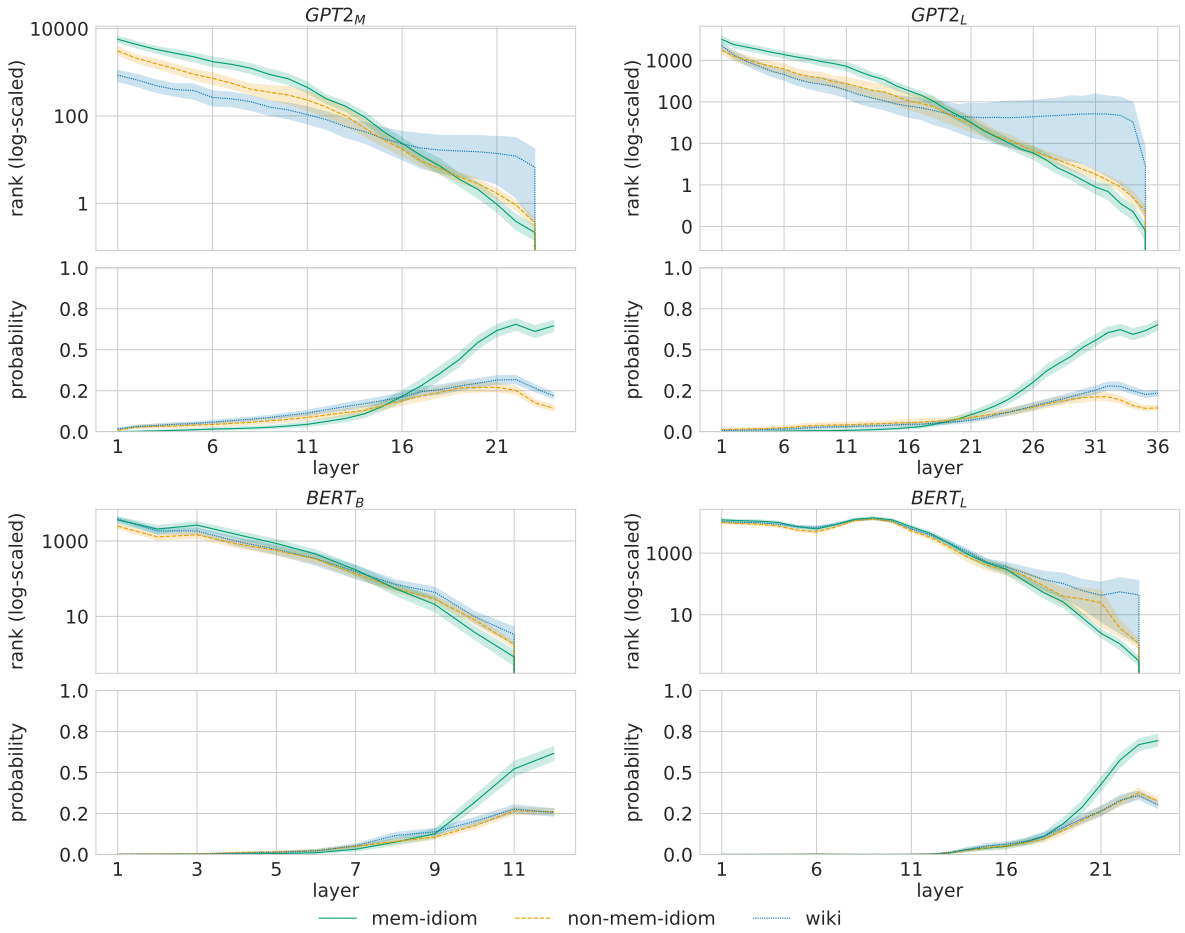

Figure 6: The predicted token's probability and log-scaled rank across layers, for memorized idioms (`mem-idiom`), non-memorized idioms (`non-mem-idiom`) and short sequences from Wikipedia (`wiki`).